

BobaLink Browser Extension “Status

Revision: 16 **Last modified:** 2026-08-20T12:27:23Z **Scope:** BobaLink (extension/) “ WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs and forwards them to the Boba merge service on port 7187. **Authority:** master plan docs/ browser_extension/IMPLEMENTATION_PLAN.md (9 phases).

Rev 16 (2026-08-20, BOB-083 /

GOVERNANCE_AUDIT_2026-08-08_ROUND2.md RD2-16)

“ staleness remediation (11.4.44/11.4.45/11.4.86):

this ledger’s last content touch was 37fcbe4 (2026-06-13, Session 12/wave-15) “ **68 days** before this pass. Real evidence gathered this session (git log, a fresh npx vitest run, npx tsc --noEmit, npm run lint “ all executed just now, not recalled from memory, 11.4.6): only **3 commits** touched extension/ in the 68-day gap “ ee4a01b (2026-06-15, a genuine MV3 service-worker-teardown resilience fix, new section below), a410b91 (2026-06-16, coverage-only test additions, no product change), and 54e313f (2026-08-08, a mechanical asyncâ€™ Promise.resolve test-fixture refactor in torrent-file.test.ts, no behavior change). This is a **TARGETED refresh** (11.4.6 option B, the same bounded-scope discipline docs/features/Status.md Rev 8 used): the new session below documents exactly what changed + a fresh full-suite run’s real result; it does **not** re-verify every pre-existing per-phase claim from Sessions 11â€™12 against the current tree (out of scope for a docs-sync pass “ see the honest gap noted in the new session block).

Captured-evidence-driven (11.4.5 / 11.4.45). Every PASS cites a real commit hash and/or a verified test/file artifact. Rows lacking runtime evidence are marked IN-PROGRESS or PENDING “ never PASS (11.4.6 no-guessing).

Baseline facts (verified this session “Session 11)

- **HEAD:** 2011810 (pushed); wave-4 (Phase-9 ci-ext.sh gate + per-store zips, ru locale, content-XSS, credential-leak audit, decrypt-throw parity, tab-group Challenge, and the _locales packaging fix) staged for the next commit.
- **Extension test corpus: 559 Vitest tests across 52 spec files “** same-session npx vitest run â€™ Tests 559 passed (559) “ under extension/tests/{unit,perf,stress,chaos,integration,security,e2e} + src/**. npx tsc --noEmit clean; npm run lint 0 errors / 0 warnings. (+126

tests over the prior 287/22 baseline: integration 7, security 4 files, stress/chaos 10, locale 4, token suites 8, tab-groups 13, ally 18, group-send nits 2, +more.)

- **WXT build wiring** **â€” COMPLETE**. WXT entrypoints exist at extension/src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html} (thin wrappers over the existing logic modules, so all prior test imports stay valid). `npx wxt build` produces a **loadable** extension/.output/chrome-mv3/ AND firefox-mv2/: `manifest_version:3`, `background.service_worker:background.js`, `popup.html`, `options.html`, `content-scripts/content.js`, `icons 16/32/48/128`, AND `_locales/{en,ru}/messages.json` (REQUIRED because `default_locale:"en" + __MSG_*__` â€” see the Â§11.4.138 note below). **Every manifest-referenced asset + the default-locale catalog verified present on disk** by the `ci-ext.sh` manual gate (Â§11.4.38). Content-script matches are derived from the curated `SITE_SELECTORS` (24 hosts, **no <all_urls>**); permissions least-privilege (no scripting/tabs/cookies; +tabGroups for Phase-5); `host_permissions = http://localhost:7187/*` only; CSP `script-src 'self'`. Per-store zips (bobalink-1.0.0-{chrome,firefox,sources}.zip) produced by `ci-ext.sh`.

Session 12 (2026-06-13) â€” test-coverage breadth + flaky-test hardening + WCAG fixes

Parallel-subagent batch (Â§11.4.103); the whole batch is verified together by `extension/ci-ext.sh` â†’ **CI-EXT: PASS**, full suite **632 passed (632)** (+73 over the 559 rc baseline), `tsc --noEmit clean`, `npm run lint 0/0`, chrome+firefox builds loadable, Â§11.4.38 asset-verify pass, both store zips â‰¥10 KiB.

- **+73 new tests across 4 files** (each anti-bluff, no absolute wall-clock thresholds):
 - `tests/perf/orchestrator-scaling.perf.test.ts` â€” machine-independent metamorphic sub-quadratic DoS-scaling guard ($t(10 \cdot N)/t(N) < 30$) + golden-good/golden-bad oracle self-validation (Â§11.4.107(10): `linearRatio=9.9` accept / `quadraticRatio=113` reject).
 - `tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts` (7) â€” orchestrator junk-flood, real BobaClient FIFO under a 500-request flood, TokenBucket rate-limit engagement, tab-group dedup-batch at scale, + chaos flaky-tab fault injection.
 - `tests/security/infohash-detection-hostile.test.ts` (32) â€” infohash hex/base32 boundary + case, xt= confusion, .torrent URL allowlist (traversal/scheme/null-byte), SHA-1 .torrent infohash correctness, dedup under hostile repetition.
 - `tests/ally/focus-and-contrast.ally.test.ts` (32) â€” focus management (WCAG 2.4.3/2.4.7/ 2.1.2) + computed colour-contrast (WCAG 1.4.3) from the REAL committed CSS, analyzer self-validated.

- **3 real WCAG AA contrast defects fixed** (src/popup/styles.css): --text-faint dark #7a7a90â†’#838399 (4.07â†’4.61), light #8a8a9câ†’#6c6c7e (3.14â†’4.76), --warning-text light #9a6a00â†’#946400 (4.40â†’4.78). REDâ†’GREEN proven; permanent guards.
- **3 flaky perf/security tests hardened** (Â§11.4.50 â€” were FAIL-bluffs under host load, proven load-coupled in isolation, then fixed to assert contention-robust intrinsic statistics): junk-flood security test (tight 5000 ms â†’ 30 s hang-ceiling + scaling moved to perf), crypto.perf (p99â†’min budget), parsers.perf (medianâ†’min scaling ratio).
- Full root-cause + evidence for all six fixes: docs/issues/fixed/BUGFIXES.md Rev 6 (entries 18â€”21).

Session 12 â€” wave-11 (2026-06-13): 6 more coverage files + 5 options-page WCAG fixes

Two further parallel-subagent waves (Â§11.4.103), verified together by extension/ci-ext.sh â†’ **CI-EXT: PASS**, full suite **761 passed (761)** (+129 over wave-10â€™s 632).

Wave-12 (2026-06-13): +38 more tests â†’ 799 passed (799) (CI-EXT: PASS): tests/security/crypto-tamper.test.ts (17 â€” AES-256-GCM tamper/auth/fixed-IV), tests/unit/link-scanner-coverage.test.ts (10 â€” scheme allowlist/visibility/dedup), tests/unit/highlight-manager.test.ts (11 â€” content-script highlight DOM no-leak/idempotency). No product defects; 1 flaky scaling-ratio test hardened (BUGFIXES 27, Â§11.4.50/Â§11.4.118).

Wave-13 (2026-06-13): +15 â†’ 814 passed (814). tests/unit/popup-states.test.ts (5) closes a genuine new gap (the popupâ€™s **partial Send-All failure** path: succeeded rows flip to Sent, failed rows stay retryable, the live region reports the failure count, Send-All re-enables). Plus **two prior-session test files brought into git** that were sitting UNTRACKED in the working tree (a working-tree-hygiene gap closed): tests/unit/options-save-flow.test.ts (4 â€” save validation/persistence/encryption) and tests/unit/offline-queue-persistence.test.ts (6 â€” cross-restart persistence/recovery), both verified real anti-bluff suites against production code. Also: a 6th flaky scaling test hardened (the tab-group **medianâ†’min** estimator, Â§11.4.50/Â§11.4.118 â€” even a median spikes under 814-test contention; min is the intrinsic-cost estimator). Autonomous test-coverage is now saturated; remaining release work is operator-gated (live-7187 round-trip, headful e2e, store assets).

Wave-14 (2026-06-13) â€” live-7187 harness prep (de-risks the #1 operator-gated blocker so it turns GREEN the moment the stack is up): the live vitest test now drives the real BobaClient + **independently confirms the synthetic torrent appears in qBittorrent** (:7186 /api/v2/torrents/info) + **cleans it up** (Â§11.4.14), all SKIP-safe; the live_detect_send_challenge.sh

had a real **11.4.14** cleanup-race defect **FIXED** (BUGFIXES 29); the HelixQA bank audited correct + in sync; and BE-1 (CORS) + BE-2 (.torrent upload) were found **already implemented** in the backend (see RELEASE_READINESS Rev 3). Default suite unchanged at **814 passed (814)** (the live test/challenge are out-of-suite, operator-run when the backend is up).

Wave-15 (2026-06-13) – remaining-blocker prep audits:
Blocker #2 (headful e2e) substantially CLEARED tests/e2e/extension-loads.spec.ts runs + passes **4/4 autonomously** (verified by a real npx playwright test run) via Playwright --headless=new: MV3 SW registers, popup + options render, and (new) the real content script auto-injects on a matched host + detects a magnet + badges it (mutation-proven). **Blocker #3 (store):** STORE_LISTING.md audited submission-ready (all required fields + permission justifications match the manifests; a real locale drift 4+8 **FIXED**); remaining = operator visual assets + submission. The 6 sibling Challenge scripts audited – all already anti-bluff correct (no changes). See RELEASE_READINESS Rev 4. Default suite still **814 passed (814)**.

- **+129 new tests across 6 files** (each anti-bluff; no absolute wall-clock thresholds):
 - tests/unit/orchestrator-dynamic-content.test.ts (5) – MutationObserver dynamic-content re-scan (insert/href-mutation/relevance-filter/debounce-coalesce/stop-disconnect), RED-proven against mutated production source.
 - tests/ally/options-contrast-motion.ally.test.ts (16) – options-page contrast + focus + reduced-motion (found 5 real WCAG defects, now fixed + guarded).
 - tests/i18n/locale-safety.test.ts (41) – i18n catalog safety across all 8 locales (placeholder/key parity, XSS-inert values, JSON integrity); tests/i18n/** wired into vitest.config.ts.
 - tests/chaos/boba-client-resilience.chaos.test.ts (9) – real BobaClient retry/backoff/ timeout/error-classification under fault injection (fake timers, ordering asserts).
 - tests/security/bencode-torrentfile-hostile.test.ts (52) – bencode parser robustness (truncation, hostile lengths, integer edge forms, deep nesting, wrong-typed info, binary-safe infohash).
 - tests/unit/text-scanner-coverage.test.ts (6) – bare-text magnet detection, cross-scanner id-equality, false-positive boundary, non-content-node skipping.
- **5 real WCAG AA options-page defects fixed** (src/options/styles.css + src/popup/styles.css; RED→GREEN, 11.4.120-reconciled guards; BUGFIXES.md 22+26): Save-button gradient 3.66+4.57:1 (button-local), two light-theme-invisible labels tokenized (1.40/1.12 + 10+16:1), reduced-motion blocks added to both stylesheets.
- **1 more flaky perf test hardened** (11.4.50/11.4.118): scanner.perf + a global vitest.config.ts setTimeout: 30s so heavy perf/stress tests are never killed by the default 5 s runner

timeout under concurrent-suite contention (their real budgets are their own internal assertions). Subagents found **no product defects** in client/bencode/text-scanner.

Session 16 (2026-08-20) – 68-day staleness catch-up (BOB-083)

Real git log -- extension/ evidence: only 3 commits touched extension/ between 37fcbe4 (2026-06-13, Session 12/wave-15) and this pass™s HEAD e0d60ab (2026-08-20). All 3 reviewed:

- **ee4a01b (2026-06-15) – genuine MV3 service-worker-teardown resilience fix (crash-audit finding S3, docs/issues/fixed/BUGFIXES.md).** tabResults was a bare top-level Map<number, PageScanResult> – MV3 service-worker in-memory state that is DISCARDED whenever Chrome idles-out and tears down the worker (~30s). After a teardown, the popup™s get-detected read and the context-menu –œSend all / send-all command both silently no-op™d (both are guarded by if (result)), so a user™s already-detected torrents vanished with no error. Fixed in extension/src/background/index.ts: every write now goes through chrome.storage.session (survives SW respawn within a browser session, cleared on browser close – the correct lifetime) via new getTabResult/setTabResult helpers, with the in-memory Map kept only as a fast-lane cache. RED (null-after-simulated-teardown) – GREEN, permanent guard extension/tests/unit/background-sw-teardown.test.ts (new file, 310 lines). Independent review (conductor-reviews-subagent, Å§11.4.70/Å§11.4.142): **GO**, no mutation residue.
- **a410b91 (2026-06-16) – coverage-only, no product change.** src/parser/torrent-file.ts coverage 75.4% – 89.9%: parseTorrentFromUrl was untested (success / 404 / network-wrap / content-type paths) and computeInfohash™s error-wrap path was untested. Every new assertion RED-proven against a mutated regression (Å§11.4.115) then restored GREEN. No defect found.
- **54e313f (2026-08-08) – mechanical test-fixture refactor, no behavior change.** tests/unit/torrent-file.test.ts™s vi.stubGlobal("fetch", –|) mocks rewritten from async () => ({–|}) to () => Promise.resolve({–|}) (same resolved shape); extension/package-lock.json touched incidentally.

Fresh evidence gathered THIS session (2026-08-20, real commands, not recalled):

- npx tsc --noEmit – clean (exit 0).
- npm run lint (eslint src/ tests/ --ext .ts) – **0 errors / 0 warnings.**
- npx vitest run (full suite, this host, HEAD e0d60ab) – **69 spec files (63 passed / 6 failed), 822 tests (815 passed / 7 failed)**, duration 256.5s. find tests src -iname '*.test.ts' -o -iname '*.spec.ts' independently counts **71** spec files on disk – the 2-file

gap is tests/e2e/extension-loads.spec.ts (Playwright, not collected by vitest run) plus tests/live/download-endpoint.live.test.ts (reachability-gated, honest SKIP when the backend is down per Â§11.4.3, not counted as a vitest pass/fail here). This raises the previously-cited “814 passed (814)” baseline (Session 12/wave-15, 2026-06-13) to a real, freshly-measured **822** total (+8, from a410b91’s new torrent-file.ts coverage tests).

- **The 7 failures are ALL bounded-time perf/stress/security-budget assertions, not functional test failures**, and this run executed on a host that was simultaneously running this docs-refresh agent’s other work (heavy concurrent load “9 node (vitest N) worker processes observed live): tests/perf/scanner.perf.test.ts (p99 budget), tests/perf/parsers.perf.test.ts, tests/stress/orchestrator-ratelimiter-tabgroup.stress.test.ts (sub-quadratic scaling ratio), and tests/security/scanner-hostile-input.test.ts (junk-flood 30s bound “observed 120.9s, “^4 over). This is the **exact same host-load-coupled flakiness class** Session 12 already found and partially hardened for sibling tests (BUGFIXES 27, Â§11.4.50/Â§11.4.118) “a tight wall-clock budget failing under contention is a known instrument limitation on this class of test, not by itself proof of a product regression.
- **PENDING FORENSICS**: whether these 7 are pure host-contention flakes (the established pattern) or a genuine new timing regression is **not resolved by this pass** “confirming that requires re-running the suite in isolation on a quiet host, which is out of this docs-only task’s charter (this agent does not own tests/ or extension/src/; see the task-scope note in the Rev 16 header above). Recorded honestly rather than asserted either way (Â§11.4.6).
- **Build artifacts**: .output/chrome-mv3/ present on disk; firefox-mv2/ and the per-store bobalink-1.0.0-*.zip files are **not** currently present in the working tree (expected “gitignored build outputs of ci-ext.sh, not regenerated in this pass since this is a docs-sync task, not a build task). ci-ext.sh itself is present and executable; not re-run here.
- **Honest scope boundary (Â§11.4.6)**: the per-phase table below and the Session-11/12 narrative above are carried forward UNCHANGED from Rev 15 except where this session’s evidence directly bears on them (the S3 resilience fix is documented here, not retrofitted into the Session-11 Phase-4/5 rows, to avoid rewriting history this pass did not re-verify). A full per-phase re-audit against the current tree was judged out of the bounded scope of this staleness-catch-up pass, exactly as docs/features/Status.md Rev 8 scoped itself.

Per-phase status

Phase Title	Status	Evidence
1 Foundation & scaffolding (WXT config, TS, shared	PASS	Scaffolded @33a9815; shared-lib + constants + type-guard unit specs green @15a9a61

Phase	Title	Status	Evidence
	libs, types, constants)		
2	Core detection & parsing engine (parsers, scanners, infohash dedup)	PASS	Parsers @7225470/fa03323; SHA-1 infohash + link/text scanners + perf/stress @fa03323; scanner orchestrator cross-scanner dedup @946c61e
3	Extension shell "content / background / popup / options"	PASS	Shell @e8fde43; background service-worker message router capstone @15a9a61
	WXT build wiring (entrypoints/, wxt build "output/, manifest validate)	PASS	Entrypoints src/entrypoints/{background.ts,content.ts,popup/index.html,options/index.html}; npx wxt build "loadable .output/chrome-mv3/ (8/8 manifest assets verified present, "11.4.38); matches derived from SITE_SELECTORS, no <all_urls> (Session 11 @5edf6ac, pushed) API leaf @e8fde43. Phase-7 decrypt-before-send wired (Session 11): BobaClient.create() decrypts the encryptedBobaApiToken bundle via shared/crypto; background reads the session passphrase from chrome.storage.session and sends the decrypted plaintext (default-open when locked). Token suites boba-client-token.test.ts 5 + background-token.test.ts 3 green. PENDING: live-7187 integration (require_backend(7187)), end-to-end detect"send"torrent-in-qBittorrent on the real backend Wired (Session 11): src/tabgroups/index.ts (dedupe across a group + batch dispatch, 13 tests) integrated into background/index.ts MENU_SEND_GROUP (deduped group batch "one addMagnets POST); manifest +tabGroups (MINIMAL "research-confirmed tabs NOT needed since only tab.id is read; "11.4.120 security-test reconciliation, mutation-verified). New background.test.ts MENU_SEND_GROUP test (RED-
4	Boba backend integration (real client, queue auth, BE-1/2/3)	IN-PROGRESS	
5	Tab-group batch (chrome.tabGroups "per-tab scan "batch send)	IN-PROGRESS	

Phase Title	Status	Evidence
6	UI/UX, i18n, accessibility, themes	IN-PROGRESS
7	Security & credentials (delegate-by-default, no embedded key, log redaction)	IN-PROGRESS

verified: handler no-op → FAIL). Independent review: **GO-with-nits**. Group-send nits (a) offline-queue enqueue-on-failure, (b) network-error notification, (c) hardened async flush → all **FIXED** (Session 11, RED→GREEN + 3—deterministic, independent review GO). **TRACKED:** decrypt-throw (wrong passphrase) on group send not enqueued/notified → pre-existing parity with Send-All, future phase

i18n: locale.test.ts (en completeness) + **ru+de+fr+es+it+pt+ja locales added** (8 total → plan target reached, 29-key parity, packaged chrome+firefox) with generalized locale-parity.test.ts (key + placeholder parity, mutation-proven); **ally**: tests/ally/{popup,options}.ally.test.ts (18 tests → roles/accessible-names/tablist→i, ↳tabpanel/live-regions, mutation-proven). tablist keyboard-nav (WAI-ARIA Arrow/Home/End). **PENDING:** deeper WCAG (contrast/focus), theme-switch evidence

crypto.ts adopted; **decrypt-and-send path landed + decrypt-throw on wrong passphrase now enqueues+notifies** (parity, RED→GREEN); session passphrase from chrome.storage.session, **no embedded key**, plaintext/passphrase never logged; security suite tests/security/* (least-privilege manifest, CSP, no-hardcoded-secret, secret-storage, **content-XSS** → render path proven safe via innerHTML-mutation→FAIL); **§11.4.10.A credential-leak audit** challenges/security/credential_leak_audit.sh (PASS, mutation-verified); **message-router robustness** (16 tests) + **isValidScanResult scan-result trust-guard + resolveTargetTabId cross-tab confused-deputy guard** (fixed a real content-script info-leak:

Phase Title	Status	Evidence
8	Testing to 100% (all types) + Challenges + HelixQA	IN-PROGRESS
9	Build, packaging & distribution (manual â€” NO CI/CD)	IN-PROGRESS

forged payload.tabId could read another tab's detections; sender.tab.id now wins) + scanner-hostile-input (19) + sender-trust (10) + content-XSS. **PENDING:** full pen-test suite (sender-origin validation, rate-limit)
 559 tests/52 files green (unit/perf/stress/chaos/integration/security/e2e). Challenge challenges/extension/
 detect_and_forward_challenge.sh drives the REAL orchestrator+client end-to-end, PASS on captured evidence, mutation-verified (no-op stub â†' FAIL). HelixQA boba-bobalink.yaml BOBA-LINK-007 (detectâ†'forward) + BOBA-LINK-008 (tab-group batch) added; symlinked into challenges/helixqa-banks/.tab-group + detectâ†'forward Challenges both mutation-verified. E2E tests/e2e/extension-loads.spec.ts is a real MV3-load test, **operator-gated SKIP** in this headless sandbox (extension load unsupported; Â§11.4.3). **PENDING:** the full 13-type coverage matrix + live-backend integration + the coverage ledger to 100%
Manual gate extension/ci-ext.sh (Session 11, Â§11.4.18 docâ€™d): tsc â†' lint â†' full vitest â†' chrome+firefox builds â†' Â§11.4.38 artifact-verify (opens the manifest, asserts every referenced asset + the default_locale catalog exist non-zero) â†' per-store wxt zip. **CI-EXT: PASS** â€” produces loadable chrome-mv3/ + firefox-mv2/ + bobalink-1.0.0-{chrome,firefox,sources}.zip. **PENDING:** store-listing metadata/submission + Â§11.4.65 user/dev/install doc siblings. NO CI/CD (manual only)

Status legend

- **PASS** “ implemented and backed by a cited commit and/or verified test/file evidence.
- **IN-PROGRESS** “ partially landed; some sub-tasks done with evidence, remainder explicitly enumerated as PENDING.
- **PENDING** “ not yet started / no runtime evidence; planned in IMPLEMENTATION_PLAN.md.

Anti-bluff notes (Â§11.4 / Â§11.4.6 / Â§11.4.69)

- The 559 figure IS a same-session recorded `npx vitest run` result (Tests 559 passed (559)), not merely a static `grep` “ it supersedes the prior 287/22 `grep`-count baseline. `tsc + lint` captured clean in the same session.
- The Â§11.4.38 loadable-artifact claim is verified by `ci-ext.sh` **opening the produced .output/chrome-mv3/manifest.json** and confirming every referenced asset AND the `default_locale` catalog exist non-zero on disk “ not by “œbuild exited 0œ.”
- **Â§11.4.138 bluff-audit (corrected this session).** An earlier Status revision claimed the artifact was loadable on an “œ8/8 referenced assets presentœ check. That check was INCOMPLETE: the manifest declares `default_locale: "en" + __MSG_*__`, but the build packaged **no _locales/** (WXT copies static assets from `src/public/`, while the catalog lived at `src/_locales/`) “ so Chrome would have REJECTED the extension at load. Caught by the new `ci-ext.sh` Â§11.4.38 step (which checks the `default_locale` “ ‘ _locales/ invariant). Root cause fixed by moving the catalog to `src/public/_locales/{en,ru}/`; the rebuilt artifact now packages it (verified on disk). Permanent guard = `ci-ext.sh` (the regression canœ™t recur silently). This is the Â§11.4.108 lesson: `build-exit-0` “ deployable-artifact.
- The Phase-7 decrypt path is proven by RED“GREEN: ciphertext-on-the-wire (pre-fix) “ decrypted-plaintext header (post-fix), with the ciphertext never sent and the token never logged (`background-token.test.ts`, `boba-client-token.test.ts`).
- The detect“forward Challenge was independently re-run and its no-op-stub negation verified (stubbed orchestrator “ FAIL) during code review.
- Independent multi-lens subagent code-review (correctness/security/build/docs) was run; the security + correctness + build + docs lenses returned GO / GO-with-nits. (The correctness/security/build subagent lenses were re-run in the conductor context after a transient platform throttle on subagent dispatch; an independent subagent re-pass runs before any release tag per Â§11.4.40.)
- **Two real bugs caught by the new tests + fixed (wave-5, RED“GREEN, independently re-verified):** (1) a `popup`“i,Žbackground CONTRACT bug the integration round-trip exposed “ the `popup` read `r.torrent.id` but the real background returns a flat `SendOutcome {id,œ},` so a SUCCESSFUL send threw in

the popup and showed a FALSE “Send failed” (row never flipped to Sent); the popup-unit fake had masked it with the wrong shape.

Fixed (popup reads `r.id`; fake corrected; integration assertion reconciled §11.4.120). (2) a scan-result TRUST bug the message-router robustness tests exposed “a hostile content script could send `{items:<non-array>}` and overwrite a tab’s good detection set with garbage; fixed with an `isValidScanResult` shape-guard (16 robustness tests; the non-array test seeds-good’s corrupt and asserts the good set survives).

- de/fr locales added (4 total, all 29-key parity, packaged in both chrome+firefox builds); orchestrator + rate-limiter stress (evidence-captured, deterministic); popup/options’ background integration round-trips (real `sendMessage`’onMessage bridge); USER/INSTALL/DEV guides (§11.4.65 exports).